Assignment 1: Needfinding

Exploring urban travel planning, group dynamics, and spontaneous navigation.

- Introduction & Domain
- Methodology & Participants
- Key Findings & Quotes
- Deep User Needs & Solutions

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Domain of Interest

Theme 3 – Smart Cities & Urban Life



Domain

Planning, organizing, and executing trips or visits to unfamiliar cities and urban areas.

Target

Young urban travelers (20–30 years old).

Project Goal

Identify unmet user needs in urban mobility to design a solution that promotes flexible, stress-free exploration.

Participants

Why: Selected frequent travelers who actively plan urban itineraries and handle group decisions.



Demographics:

- 8 participants (ID1-ID8)
- ages 22 to 30
- 3 European, 4 Asian, 1 African
- balanced gender mix

Extreme Users:

- ID4 (***Extreme Planner***): 50k steps/day, AI-generated map code
- ID3 (***Privacy Extreme***): Rejects local apps; relies strictly on human advice.

Domain Experts:

- ID1, ID5: Both have an academic background in tourism and translation.

Interview Setup

How, where, and what we observed during the interview

01

Where

- University study rooms
- participants' homes
- a restaurant
- online videocalls

02

What We Asked

- Planning: Habits and app switching.
- Group Choices: Budget, time and preferences.
- Flexibility: Unexpected changes and queues

03

Team Roles

Managed both roles in every session:

- leading the interview
- taking notes/audio recordings.

04

Materials Used

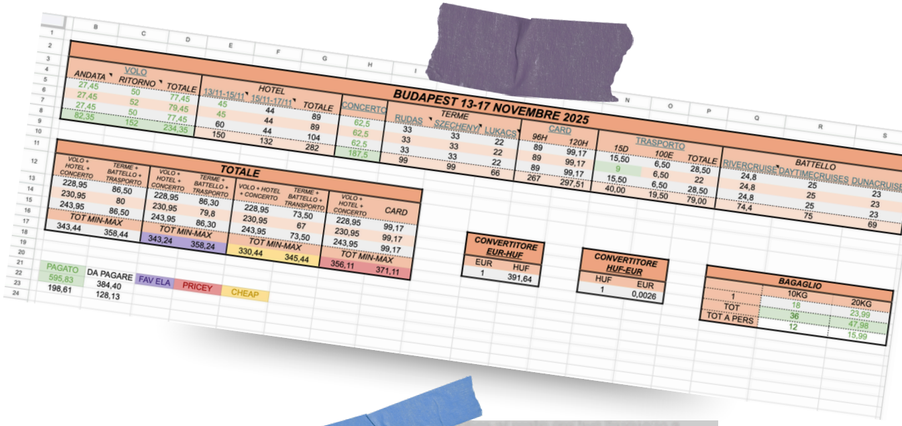
Tablet (for audio recording & taking notes)

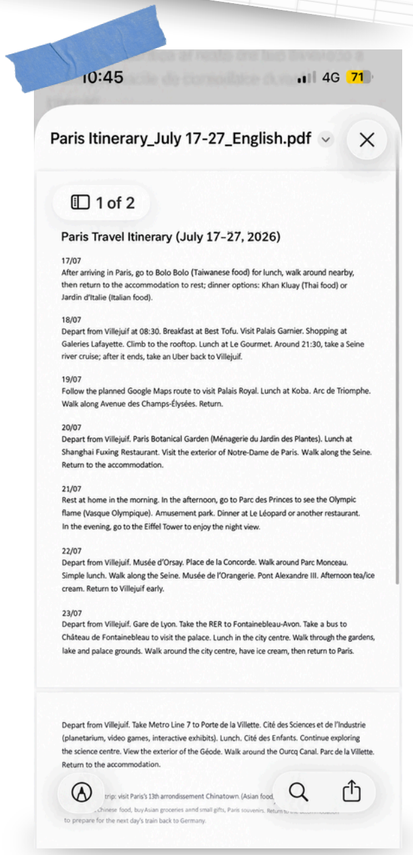
Key Observation

High cognitive friction observed when participants manually switch between Excel, Google Maps, Notes, and AI chatbots during planning.

Key Findings & User Quotes

Tools used by participants and quotes highlighting their planning pains.





EXCEL & TABLES

Used by extreme planners (e.g. ID4) to organize timelines, track budgets, and calculate walking steps.

NOTES & CHATS

Used to paste links, save addresses, and share ideas in WhatsApp/Telegram group chats.

MAPS & AI

Used to visualize routes and generate quick suggestions, causing frequent app-switching.

KEY QUOTE #1

“I use Excel to track places, reviews, and costs, but then I have to switch to Google Maps to check distances between them.”

KEY QUOTE #2

“To create an itinerary, I need 20 tabs open on my computer to connect places, and then I manually write the final plan in my phone notes.”

User Needs

28 specific **needs** identified across 8 interviews, **categorized** into **4 deep needs**.



TRAVEL INFORMATION INTEGRATION

Needs 1, 2, 3, 10, 14, 15, 26:
Centralize transport, costs, maps, and schedules without manual copy-pasting.



ELASTIC ITINERARIES

Needs 4, 8, 12, 13, 21, 23, 24:
Balance group desires and quickly adapt to weather, delays, or tiredness.



AUTHENTIC LOCAL INFO

Needs 5, 9, 11, 19, 25, 28:
Access real visual proofs and local advice away from tourist traps.



FLUID MOBILITY

Needs 6, 7, 17, 18, 20, 22, 27:
Skip lines and get transit guides without installing extra local apps.

NEED COUNT



Organizers focus on tool consolidation (14 needs), while Explorers prioritize flexibility and authenticity (14 needs)

Proposed Solution

An intelligent urban platform offering real-time, context-aware alternative routes to solve travel friction.

All-in-One Hub

Centralizes itinerary planning, expenses, and map routes to eliminate app-switching.

SOLVE DEEP NEEDS: A, D

Elastic Scheduling

Dynamically recalculates routes for weather, queues, or group fatigue.

SOLVE DEEP NEEDS: B, D

Authentic Local Insights

Uses raw photos and verified resident advice to bypass sponsored tourist traps.

SOLVE DEEP NEEDS: C, D

Smart Alternatives

Recommends nearby spots by time, budget, and interest.

SOLVE DEEP NEEDS: A, B, C, D

Selected Solution

Solution Overview

An integrated platform that unifies itinerary planning and real-time navigation.

- **All-in-One Hub:** Centralizes maps, budgets, and notes to eliminate app-switching.
- **Dynamic Rerouting:** Suggests nearby alternatives when facing bad weather, long queues, or fatigue.
- **Intent-Driven:** Matches your original goals (time, budget, and activity type) without showing random spots.

Motivation #1

Cognitive Load Reduction

Removes the stress of manual searching during unexpected changes by offering instant, ready-to-use options.

Motivation #2

Intent Preservation

Recommends relevant alternatives that match the original travel goal (e.g., culture, relax, local food) instead of showing random places.

Motivation #3

Spontaneity with Structure

Combines the clear organization required by Organizers with the flexibility desired by Explorers.

PROJECT IDENTITY

Project Name & Value Proposition



- How we land the project name?

“Plan B” Concept

Initially designed as a simple backup solution when unexpected delays or changes happen on the spot.

Scope Evolution

Expanded to include full itinerary creation from scratch, combining planning with spontaneous exploration.

The “*Bee*” Metaphor

Like bees buzzing around flowers, travelers can freely wander the city and discover fun local spots.

- One-line value proposition

> *“PlanBee helps travelers organize seamless itineraries and effortlessly adapt on the spot when plans change.”*